

Positional Semantics for Retrieved Native-KV Memory: RoPE, Fragmentation, and Bounded Relocation in Progressive Retrieval Attention

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Abstract

Retrieved native-key/value (K/V) memory reuses attention states that were encoded outside the current forward pass. This raises two questions that ordinary prefix caching can avoid: which source position does each cached key represent, and which causal context shaped it? We give Progressive Retrieval Attention (PRA) an explicit positional contract while preserving learned absolute position embeddings as the default and adding sinusoidal absolute and rotary position embedding (RoPE) modes. Across five matched seeds, two model capacities, and all three mechanisms, a compact logical start offset per contiguous chunk removes layer-0 position-reset error exactly and lowers final-layer K error in all 30 paired comparisons. This result survives a continuously advancing 192-token prompt whose model operations never exceed 24 tokens. Natural-language and retrieved-memory validation qualify the result. On 8L WikiText-2 windows, offsets lower final-layer K error in 30/30 pairs but lower next-token loss in only 5/30. On controlled HotpotQA- and QASPER-derived answer-code probes, routed loss improves in 35/60 pairs: learned-absolute and sinusoidal means improve, whereas RoPE means worsen in both datasets and capacities. A ranking retrospective reinforces the separation: with one of four references selected, small-tier sinusoidal target recall is 0.779/0.800 on the two probes, versus 0.454/0.408 for RoPE. Thus coordinate fidelity does not by itself guarantee useful prediction. Contextualization and routing remain separate factors, while oracle anomalies suggest memory composition as a fourth. A four-cell pre/post-RoPE matrix also establishes exact parity at matched effective positions; deferred rotation permits rebinding but raises the measured unfused warm CUDA path from 0.179 to 0.649 ms. The evidence supports source-relative positional continuity as a representation-fidelity contract for chunked native-KV transport and RoPE as an explicit rebinding algebra, not a universal long-context quality advantage.

1 Introduction

A transformer key is not only a content representation. Depending on the positional mechanism, it carries or interacts with a coordinate. This is easy to ignore while every key is produced in one dense forward pass. It becomes operational when a system encodes a long source in bounded blocks, stores native per-layer K/V, retrieves a sparse subset, and combines it with a later query. A source key may retain its document coordinate, restart at zero in each block, or receive a new coordinate when retrieved. These choices are not equivalent.

PRA provides a controlled setting for studying the problem. It stores full layer-native K/V behind inexpensive routing gists and inserts selected states before direct K/V under one softmax. Paper 1 established native transport, exact routing, model-bounded encoding, and an implicit

prompt-head reference. The present paper holds those mechanisms fixed and changes the positional semantics.

This paper asks a narrower question than whether a model can extrapolate to a longer dense context: can independently produced native K/V retain an auditable position identity, and what errors remain after that identity is repaired? Our contributions are:

- a reusable position strategy supporting learned absolute, sinusoidal, and RoPE semantics without changing the default absolute checkpoint layout;
- a weight-preserving RoPE SelfAttention-to-PRA conversion whose no-memory logits match the source model;
- compact source-relative offsets, explicit position metadata, and pre/post-position state labels on cached native K/V;
- probes separating position reset, contextual block fragmentation, overlap, pre- versus post-RoPE K, retrieved-distance policies, and continuously advancing implicit-head behavior;
- a four-phase validation across controlled synthetic probes, WikiText-2, and HotpotQA/QASPER-derived retrieved-memory tasks; and
- five-seed tiny/small evidence in all three position modes, including negative task-loss, routing, composition, and variance results that constrain the claims.

2 Progressive Retrieval Attention in Brief

PRA keeps a recent direct tail continuously visible and makes older or external material addressable. A long source is first encoded in model-safe blocks. Each layer publishes its native K/V, and those states are sliced into smaller routing chunks. A gist is a compact routing summary of a chunk; it is *not* the chunk’s K/V and is never used as a substitute attention value. At query time, gist scores select chunks, the selected chunks’ full token-level native K/V are materialized, and memory K/V attend beside direct K/V under one softmax.

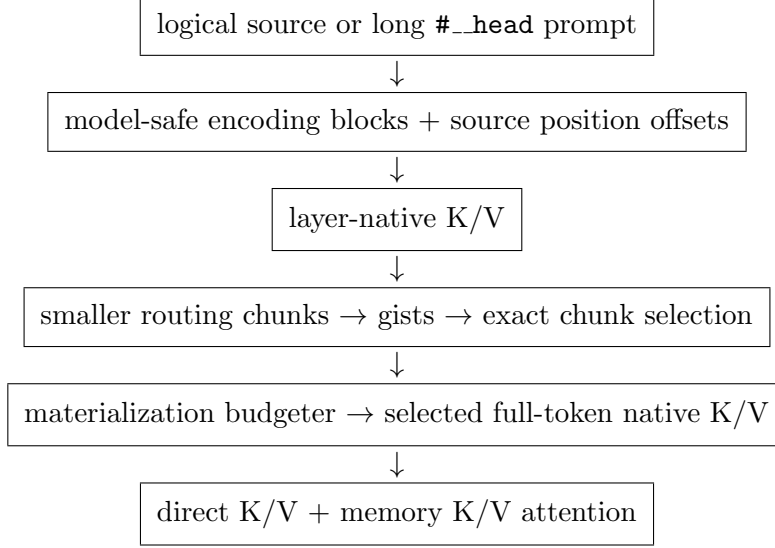


Figure 1: PRA data path. Gists choose chunks; attention consumes the selected chunks’ native token-level K/V. Position continuity is established during bounded encoding and audited again at materialization.

Four lengths describe different constraints:

$$L_{\text{logical}} \gg L_{\text{model_op}} \geq L_{\text{encode}} > L_{\text{route}}, \quad L_{\text{direct}} + L_{\text{materialized}} + L_{\text{reserve}} \leq L_{\text{model_op}}. \quad (1)$$

L_{logical} is the source coordinate span. $L_{\text{model_op}}$ is the maximum number of tokens in one model invocation. L_{position} , introduced below, is the largest representable coordinate and need not equal the operation limit. L_{encode} bounds contextual cache construction, while L_{route} controls retrieval granularity. A long logical source therefore does not imply a long dense model call.

2.1 A concrete position-reset failure

Consider tokens A–H at logical positions 0–7, encoded as A–D and E–H. If the second block restarts at zero, E receives the coordinate formerly assigned to A. Source-relative encoding instead uses

$$p_{\text{logical}}(j) = p_{\text{chunk,start}} + j. \quad (2)$$

Source token	A	B	C	D	E	F	G	H
Source position	0	1	2	3	4	5	6	7
Reset chunk position	0	1	2	3	0	1	2	3
Offset chunk position	0	1	2	3	4	5	6	7

Thus token 37 within a block beginning at 4096 receives position 4133. A contiguous routing chunk needs one integer start offset, not a position tensor per token; explicit tensors remain useful for irregular layouts and audit output.

For a continuously advancing prompt head, rollover preserves one coordinate system as direct tokens become historical memory. Independent URI references are different: each source owns its own coordinate system unless an application defines a shared layout. The implementation therefore treats source identity and source-relative offset as cache provenance rather than silently concatenating unrelated documents into one global axis.

3 Background and Related Work

Learned absolute embeddings add a position-dependent vector before the decoder stack and therefore entangle content and position in later hidden states. Transformer-XL introduced relative positional treatment for recurrent segment memory [2]. Shortformer moved absolute position information to queries and keys to improve cached-state reuse across subsequences [8]. ALiBi adds distance-dependent attention biases for train-short, test-long extrapolation [9].

RoPE rotates query and key feature pairs by position-dependent phases, making their dot product depend on relative displacement while each vector retains an absolute phase [10]. Position Interpolation rescales indices into a trained range [1]; LongRoPE uses nonuniform interpolation and progressive extension [4]. Those methods enlarge dense model context. Our question is different: how should positions behave when only selected externally stored native K/V is returned? StreamingLLM preserves selected chronological cache states and attention sinks [11]; PRA retrieves URI- or request-addressed chunks, so it must also specify source positions, relocation, and routing semantics.

Cache reuse systems expose adjacent parts of the same problem. Prompt Cache reuses modular attention states while preserving a position-aware prompt schema [5]. CacheBlend observes that independently cached non-prefix chunks miss preceding context and selectively recomputes states [14]. MiniPIC stores unrotated keys and binds RoPE positions inside attention [7]; SemPIC instead learns a cache writer to reduce residual context dependence [12]. These approaches motivate the same distinction tested here: correcting coordinates does not reconstruct absent causal context, and flexible position binding does not by itself select useful memory.

4 Positional Semantics for Native K/V

4.1 What RoPE invariance does and does not say

For one head of width d_h , let R_p be the block-diagonal rotation at position p :

$$\tilde{q}_i = R_i q_i, \quad \tilde{k}_j = R_j k_j. \quad (3)$$

Orthogonality and the group property give

$$\tilde{q}_i^\top \tilde{k}_j = q_i^\top R_i^\top R_j k_j = q_i^\top R_{j-i} k_j. \quad (4)$$

Thus translating both positions by c leaves the score unchanged: $(R_{i+c} q_i)^\top (R_{j+c} k_j) = q_i^\top R_{j-i} k_j$. This supports relocation of a complete query–memory relation. It does not imply that moving only a cached key leaves attention unchanged; a fixed query should observe the changed relative distance.

4.2 Three independent choices

Three choices are often collapsed into "RoPE support":

1. **Encoding positions:** local block indices or logical source indices.
2. **Storage state:** pre-positional K_{raw} plus metadata, or published post-RoPE K_{rope} .
3. **Materialization policy:** preserve source distance, reset locally, relocate as a block, compress distance, or retain order only.

The canonical implementation publishes post-position K. Pre-position storage remains experimental because deferred rotation changes the cache representation and runtime work.

4.3 Position and contextualization fragmentation

If a dense source is split into bounded blocks, local position reset changes absolute embeddings and post-RoPE key phases. Independently, later blocks lose causal left context, which changes hidden states even with perfect position IDs. We call these *positional fragmentation* and *contextualization fragmentation*. Logical global positions can remove the first at layer 0. Encoding overlap can only approximate the second.

Table 1: Four primary factors and two RoPE storage-policy extensions that must be measured separately.

Failure or factor	Cause	Diagnostic	Control or implication
Positional fragmentation	Chunk coordinates reset	Layer-0 K mismatch	Source-relative off-sets
Contextualization fragmentation	Earlier causal context is absent	Deeper mismatch after offset repair	Overlap or recomputation
Selection failure	Router misses useful chunks	Routed-oracle gap and gist recall	Better routing
Memory composition	Relevant chunks may omit support, interact poorly, or add distractors	Evidence-only set is not always best	Treat utility as a set-composition problem
Positional rigidity	Phase is baked into K	Deliberate rebinding probe	Raw K plus deferred RoPE
Remote approximation	Relative phase is compressed	Exact-distance output deviation	Policy-specific adaptation

Selection asks which chunks are chosen. Composition asks which combination of chosen chunks is beneficial to the model’s current prediction. Routed-oracle gaps directly support selection as an independent limitation. Composition is a candidate fourth factor: the observed oracle anomaly suggests it, but does not establish a universal composition law. In particular, a known evidence chunk is not automatically a performance oracle.

5 Implementation

The shared code adds a registry-backed `PositionEncoding` interface. Its first method `positions` token embeddings; its second transforms per-head Q/K with shape $[B, H, T, d_h]$. Absolute mode adds the existing learned table and leaves projected Q/K unchanged. RoPE mode leaves token embeddings unchanged and rotates Q/K. A position-aware SelfAttention block is instantiated only for non-absolute modes, so old absolute models retain their original `TransformerEncoderLayer` state names and behavior.

Table 2: Position mechanisms under retrieved native-K/V reuse.

Mechanism	Source offsets	Coordinate range	Retrieval-time re-binding
Learned absolute	Added before layer 0	Limited by learned table L_{position}	Not separable from hidden state
Sinusoidal absolute	Added before layer 0	Formula-defined; extrapolation unverified	Not separable from hidden state
RoPE, post-position K	Applied to Q/K	Formula-defined; extrapolation unverified	No, without inverse rotation
RoPE, pre-position K	Applied when consumed	Formula-defined; extrapolation unverified	Yes, by choosing effective IDs

The configuration separates `model_max_context_tokens`, the native operation limit, from `position_capacity`. Learned absolute IDs are checked against the latter. This allows, for example, a 32-token bounded operation to use positions 160–191 in a model with a 192-entry position table. RoPE accepts larger integer IDs, but acceptance by the formula is not evidence that a model trained only on short distances will use them well.

PRA attention applies the same strategy in direct and cache-publication paths:

```

q = split_heads(q_proj(hidden))      # [B,H,T,Dh]
k = split_heads(k_proj(hidden))      # [B,H,T,Dh]
q, k = position.transform_qk(q, k, position_ids)

k_ref = split_heads(k_proj(reference_hidden))
_, k_ref = position.transform_qk(k_ref, k_ref, reference_positions)
cache.publish(k_ref, v_ref, position_ids, state="post_position")

```

Values are not rotated. Each `LayerKV` records the positions used to publish K and a `position_state`. `ReferenceChunk` and `ReferenceChunkMemory` carry `logical_start` and `logical_end`; validation, overlap, overflow splitting, routing selection, CPU residency, and materialization preserve them. A contiguous chunk can reconstruct token IDs as `logical_start + arange(length)`. The result trace exposes the same range so a consumer can audit which native states entered attention.

Conversion copies Q/K/V projections, output projection, norms, feed-forward layers, embeddings, and language-model head from trained RoPE SelfAttention into PRA. With memory disabled, source and converted logits match to unit-test tolerance. No memory adapter or additional transport projection is introduced.

For a long prompt, the adapter keeps a direct tail and registers displaced tokens under `#_head`. Bounded encoding produces native K/V, smaller chunks produce routing gists, and whole chunks are selected under

$$T_{\text{direct}} + T_{\text{memory}} + T_{\text{reserve}} \leq L_{\text{model_op}}. \quad (5)$$

RoPE can continue logical IDs past $L_{\text{model_op}}$ without enlarging an operation. Learned absolute mode can do the same only while IDs remain below L_{position} .

6 Experimental Design

The study proceeds in four phases. Phase I uses tiny controlled models as a mechanism microscope for translation, storage state, rebinding, distance policy, and split scaling. Phase II asks whether the headline reset-versus-offset effect survives the small tier and the sinusoidal control. Phase III tests bounded next-token prediction on contiguous WikiText-2. Phase IV tests PRA in controlled reference-conditioned tasks derived from HotpotQA and QASPER. This ordering separates discovery from validation and prevents a successful geometric probe from being presented as end-to-end retrieval evidence.

6.1 Models and training

Controlled tiers are smaller than repository deployment profiles so all paired seeds fit one NVIDIA GTX 950M. All modes retain the learned position table, unused by sinusoidal and RoPE, to match parameter count exactly within a dataset and tier.

Table 3: Controlled matched model tiers.

Tier	Layers	Heads	d	d_{ff}	Parameters	Steps
Tiny	2	4	64	128	92,288	100
Small	4	4	128	256	580,608	150

Seeds are $\{1, 7, 21, 42, 87\}$. Each learned-absolute/sinusoidal/RoPE triple uses the same examples, optimizer, batches, vocabulary, native training context, and training-token budget within its validation family. Position mechanism is the intended difference. The synthetic matrix uses 2,048 fixed-target examples and a 192-token native context. Every run reaches 100% held-out answer-token accuracy. The original split-scaling set has 12 invariant source/query/answer examples; only source boundaries change. We evaluate 2, 5, 16, 32, and 64 splits.

PRA uses post-position K, local positions per independent reference, a 160-token whole-chunk budget, and a four-token reserve in the split experiment. Conditions include dense full context, tail only, native all, evidence oracle, gist-routed, shuffled, and memory-disabled. Recovered context benefit is

$$\text{RCB} = \frac{\mathcal{L}_{\text{tail}} - \mathcal{L}_{\text{PRA}}}{\mathcal{L}_{\text{tail}} - \mathcal{L}_{\text{full}}}, \quad (6)$$

reported without clipping.

6.2 Phase I—mechanistic discovery

The pure probe fixes Q/K/V across shifts up to 8,192 and measures logit RMSE, Jensen–Shannon divergence, top-token agreement, and output RMSE. The pre/post probe compares stored post-RoPE K with deferred rotation at identical positions, K-only relocation against a fixed query, and common Q/K translation.

The raw-K policy probe holds Q/K/V content fixed while source–query distance grows from 1L to 32L, where $L = 192$. Exact logical, immediately preceding local, 192-token clipped, log-compressed, four-level bucketed, and remote-past assignments move each 32-token chunk as a unit, preserving local order. This is a positional sensitivity probe, not downstream quality. For systems cost, synchronized warm CUDA measurements separately time selected-key gather or gather-plus-rotation, attention, and their combined path over 500 iterations for each of five seeds.

The source-continuity decomposition compares dense 128-token encoding with 32-token blocks under local reset, source-relative offsets, and 25% or 50% left overlap. It reports K RMSE by layer and paired seed. The absolute table has 192 entries, so all logical IDs are representable even though each encoding operation is bounded to 32 tokens.

The RoPE storage matrix crosses post- versus pre-position K with reset versus source-relative coordinates: A is post/reset, B post/offset, C pre/reset, and D pre/offset. B is the reference. A separate raw-K condition rebinds the source by +512. Runtime records gather, compact position reconstruction, RoPE transform, materialization, attention, and the warm combined path. All timings synchronize CUDA around 500 iterations and use five seeds.

The continuous-head probe uses $L_{\text{logical}} = L_{\text{position}} = 192$ and $L_{\text{model_op}} = 32$. It keeps 8 direct tokens, encodes 16-token blocks, routes 8-token chunks, and materializes at most 24 memory tokens. Four examples and five seeds progress from reset routing to offset routing, offset plus 50% overlap, and a constructed oracle containing the first evidence chunk. This oracle tests one selection hypothesis; it is not ground-truth proof that no other chunk helps. The earlier distance-scaling head probe is retained separately.

Before result inspection we recorded the following expected directions: offsets should reduce reset error; overlap should reduce the remaining contextual error; matched pre/post storage should agree; intentional K-only rebinding should change attention; and the constructed oracle should be no worse than routed selection. We report failed and partial expectations rather than discarding them.

6.3 Phase II—capacity and sinusoidal validation

The headline positional experiment crosses three mechanisms, two tiers, and five seeds. It compares 32-token independent blocks with reset coordinates, source-relative coordinates, and 25% or 50% left overlap against a dense 128-token comparator. The learned-absolute table has 192 entries, so this phase includes $L_{\text{logical}} > L_{\text{model_op}}$ while remaining within L_{position} . We report layer-0 and final-layer K RMSE, paired per-seed changes, and token processing cost.

The capacity head probe extends the earlier continuous-head experiment to sinusoidal positions. It uses $L_{\text{logical}} = 192$ and $L_{\text{model_op}} = 32$, keeps 8 direct tokens, encodes 16-token blocks, routes 8-token chunks, and materializes at most 24 memory tokens. The staged comparison is reset, logical offset, offset plus 50% overlap, and a constructed first-evidence- chunk oracle. Every native operation and every violation count is recorded.

6.4 Phase III—natural-language validation

We train from-scratch next-token models on WikiText-2 raw text [6] with a shared 2,000-token BPE vocabulary. Training uses 128-token sequences; evaluation uses contiguous 32-, 64-, 128-, and 256-token windows, denoted 1L, 2L, 4L, and 8L for the 32-token operation limit. Each window is evaluated as reset blocks, source-relative blocks, and source-relative blocks with 16-token left overlap. The dense window is a representation comparator. Metrics are loss, perplexity, token accuracy, layer-0 and final-layer K RMSE, logical/native ratio, largest native operation, and overlap cost. Formula-defined sinusoidal and RoPE positions can represent all tested IDs; learned-absolute IDs remain within its 256-entry table.

6.5 Phase IV—retrieved-memory validation

HotpotQA [13] and QASPER [3] supply natural source text and annotated evidence, but the evaluation is deliberately a controlled transport probe rather than unrestricted QA. For balanced

yes/no examples, the source contains an explicit `AnswerCode yes|no` anchor, dataset evidence, and sampled in-document distractors. The visible tail asks for the answer code. The five-part condition places the source into grouped, one-reference encoding blocks and compares reset, source-relative, and source-relative plus 25% overlap. Each tier/mechanism/seed is tested with routed, evidence-only, all-memory, shuffled, disabled-memory, full-context, and tail-only conditions.

Recovered context benefit (RCB) normalizes the loss reduction from tail-only toward full context. Answer-code accuracy measures exact prediction in the two-token answer space. Routing recall asks whether a selected set contains the annotated target; mean reciprocal rank (MRR) records its rank, and target coverage reports whether the top-two selection covers it. We also record selected chunks, physical K/V tokens, logical/native ratio, maximum native operation, and overlap duplication. Dense/full access is a control, not an assumed oracle: distractor competition and the training distribution can make an evidence-only or all-memory set worse than a router-selected combination.

7 Results

7.1 Phase I—mechanistic discovery

7.1.1 Pure translation and storage

Across head dimensions 8–64, common translation preserves the top-attended token exactly. Maximum logit RMSE is 5.79×10^{-5} , maximum JS divergence is 3.00×10^{-7} , and maximum output RMSE is 1.80×10^{-5} (Figure 2).

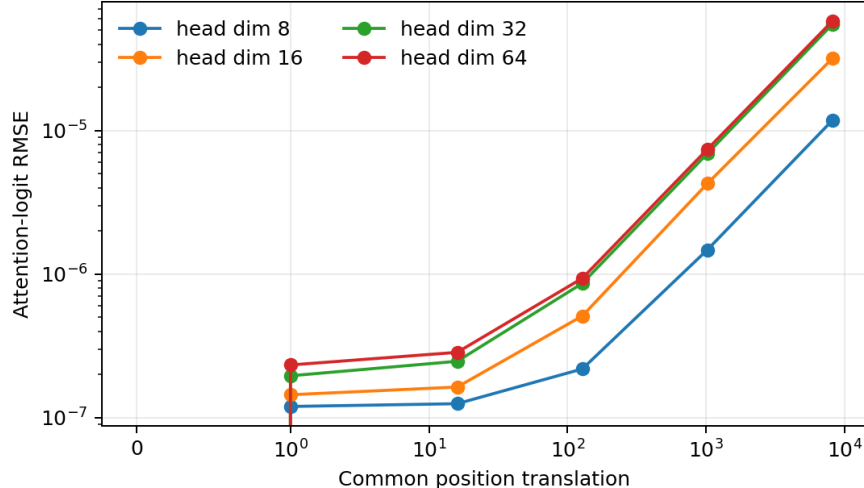


Figure 2: Numerical RoPE common-translation invariance. The logarithmic vertical axis exposes floating-point error, not a quality effect.

The trained-model storage matrix gives the same result at the model output (Figure 3). Post/reset A equals pre/reset C exactly; post/offset B equals pre/offset D exactly. B and D reproduce the offset reference with zero output RMSE and top-token agreement 1. Reset A/C instead have mean K RMSE 0.648, output RMSE 0.063, and top-token agreement 0.225. Rebinding raw K by +512 gives output RMSE 0.093 and agreement 0.05. The matrix rejects two possible

misconceptions: raw K is not intrinsically more accurate when effective IDs match, and changing K coordinates alone is not an invariant operation.

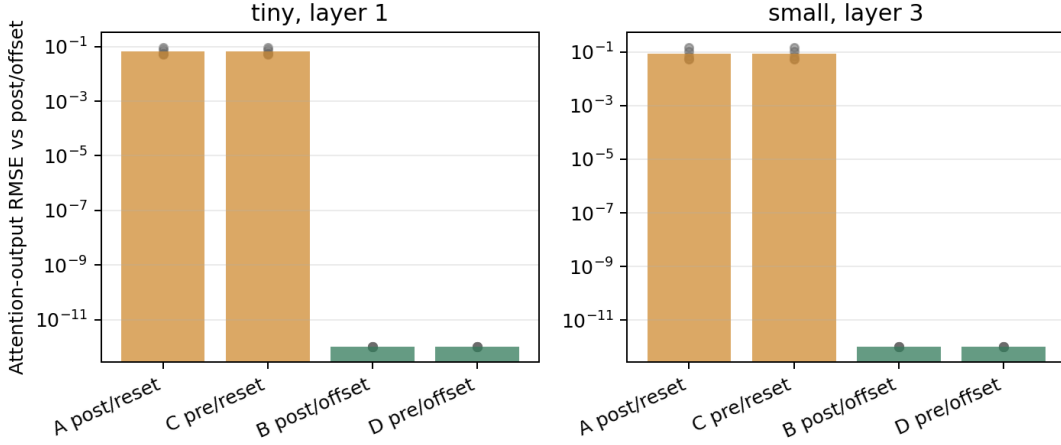


Figure 3: Four-cell RoPE storage matrix over both tiers, all layers, and five seeds. Deferred rotation is exact at matched effective positions; reset coordinates and intentional rebinding produce the expected attention changes.

The earlier tensor probe agrees: moving K alone by 128–8,192 positions changes fixed-query scores, while translating Q and K together remains near invariant (maximum RMSE 4.91×10^{-4} at 8,192). Storage state therefore controls when positions can be bound, not whether the same binding is accurate.

7.1.2 Deferred rotation and distance precision

Exact logical assignment reproduces post-position K, scores, and outputs. The approximate policies are not numerically neutral. At 32L in the small tensor tier, their mean output RMSE against exact assignment spans 0.289–0.326 and top-token agreement is 0–0.05. Tiny-tier output RMSE spans 0.259–0.290 and agreement is 0.05–0.15. These are five-seed random-content attention probes: they show geometric sensitivity, not task-loss degradation. A compressed policy therefore requires training or adaptation evidence before deployment.

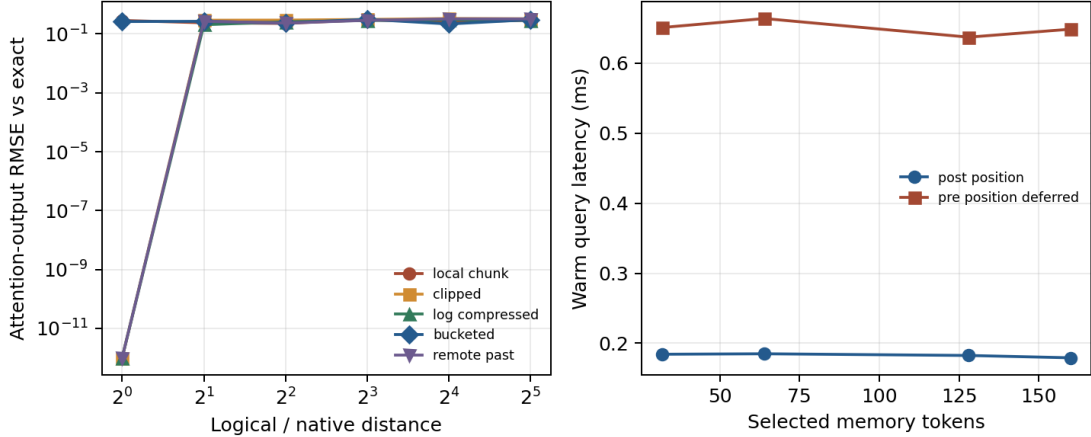


Figure 4: Left: positional-policy deviation from exact logical assignment for the small tensor tier. Right: synchronized warm CUDA microbenchmark. Approximate distance is not a no-op, and deferred rotation adds a launch-dominated operation on this GPU.

Table 4: Small-tier CUDA timing in ms, mean over five seeded runs of 500 warm iterations.

Storage	K/V	Gather	Reconstruct	RoPE	Attention	Warm
Post-position	32	.0312	.0000	.0000	.1343	.1842
Pre-position + deferred	32	.0337	.0191	.4021	.1343	.6511
Post-position	160	.0327	.0000	.0000	.1342	.1791
Pre-position + deferred	160	.0329	.0145	.3838	.1342	.6487

For 192 stored small-tier tokens, both modes use 196,608 bytes of fp32 K/V. The auditable cache can reconstruct a contiguous position range from an 8-byte start offset per chunk; a full 192-token int64 audit tensor would cost 1,536 bytes (0.781%). Selecting 160 tokens materializes 163,840 K/V bytes. Raw K changes when phase can be assigned, not how many K/V elements are retained. Position reconstruction is small; the unfused RoPE transform dominates prototype overhead. The nearly flat token-count timing reflects kernel-launch dominance on a GTX 950M, so this experiment supports no production throughput claim.

7.1.3 Split scaling

Figure 5 reports the trained experiment. Split 2 often creates a chunk too large for the 160-token whole-chunk budget, so native-all becomes tail-only. This is a materialization outcome, not a positional failure.

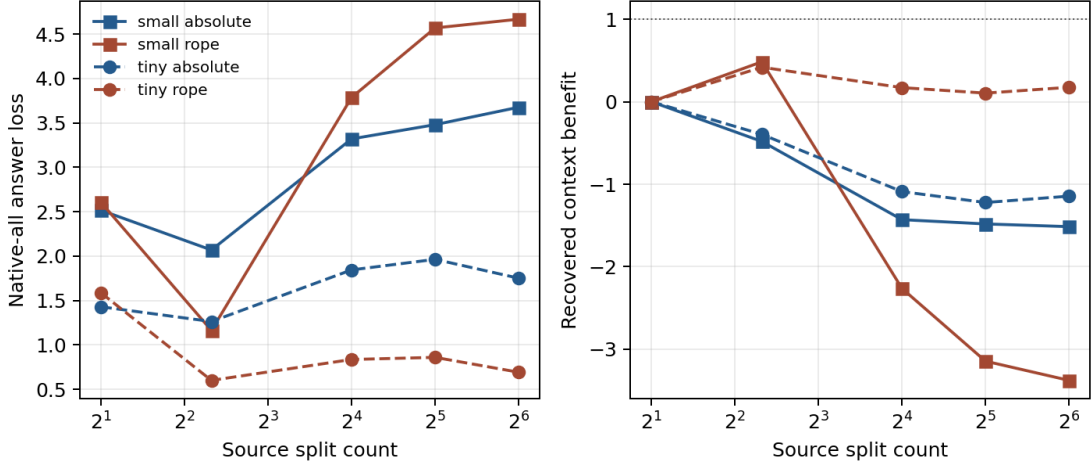


Figure 5: Five-seed native-all loss and recovered context benefit. Negative RCB means that unfiltered memory hurts relative to tail-only inference.

Table 5: Mean native-all loss by split count. Lower is better.

Tier	Position	5	16	32	64
Tiny	Absolute	1.264	1.845	1.964	1.752
Tiny	RoPE	0.601	0.837	0.860	0.693
Small	Absolute	2.069	3.321	3.481	3.677
Small	RoPE	1.163	3.787	4.570	4.669

Tiny RoPE is lower in all five paired seeds at 16 and 32 splits. With five seeds, however, the smallest exact two-sided sign probability is 0.0625. At 64 splits it is lower in four of five. The small tier reverses in mean at 16–64 and has high seed variance. There is no scale-independent RoPE advantage.

Oracle selection isolates transport and sparsity from routing. At 64 splits, tiny absolute/RoPE oracle losses are 0.581/0.070, corresponding to RCB 0.483/0.949; RoPE is lower in all five seeds. Small oracle losses are both low, 0.126/0.046, with RCB 0.967/0.929. Routed losses remain much larger (tiny 1.994/1.301; small 4.434/4.721). Strong oracle outcomes beside weak routed outcomes locate much of the remaining failure in selection, not transport.

7.2 Phase II—capacity and sinusoidal validation

7.2.1 Source-relative continuity and overlap

Source-relative offsets make layer-0 K exactly match the dense 128-token comparator in every tier, mechanism, and seed: 30/30 paired runs reach zero RMSE. Reset layer-0 means range from 0.3345 to 0.6673 (Table 6). Sinusoidal positions show the same exact repair as learned absolute positions and RoPE, falsifying the narrower explanation that the effect is a special consequence of rotary algebra.

Deeper layers retain contextualization fragmentation. Offsets lower final-layer RMSE in all 30 pairs, but the residual is larger in the small tier for every mechanism. The intervention therefore

survives the sixfold parameter increase without becoming a complete dense-state reconstruction. Twenty-five-percent overlap lowers the final-layer mean in all six conditions. Fifty-percent overlap lowers all three tiny means and the small sinusoidal/RoPE means, but slightly raises small learned-absolute RMSE from 0.3066 to 0.3103. Overlap is a historical- context approximation with additional encoding cost, not a monotonic coordinate repair.

Table 6: Three-way reset-versus-offset validation. Values are means over five seeds; every final-layer paired difference favors source-relative offsets.

Tier	Position	Reset L0	Offset L0	Reset final	Offset final	Seeds improved
Tiny	Learned absolute	.6673	.0000	.6769	.1129	5/5
Tiny	Sinusoidal	.3640	.0000	.4722	.1436	5/5
Tiny	RoPE	.6463	.0000	.6745	.1373	5/5
Small	Learned absolute	.6626	.0000	.7193	.3066	5/5
Small	Sinusoidal	.3345	.0000	.5507	.2746	5/5
Small	RoPE	.5586	.0000	.7498	.4235	5/5

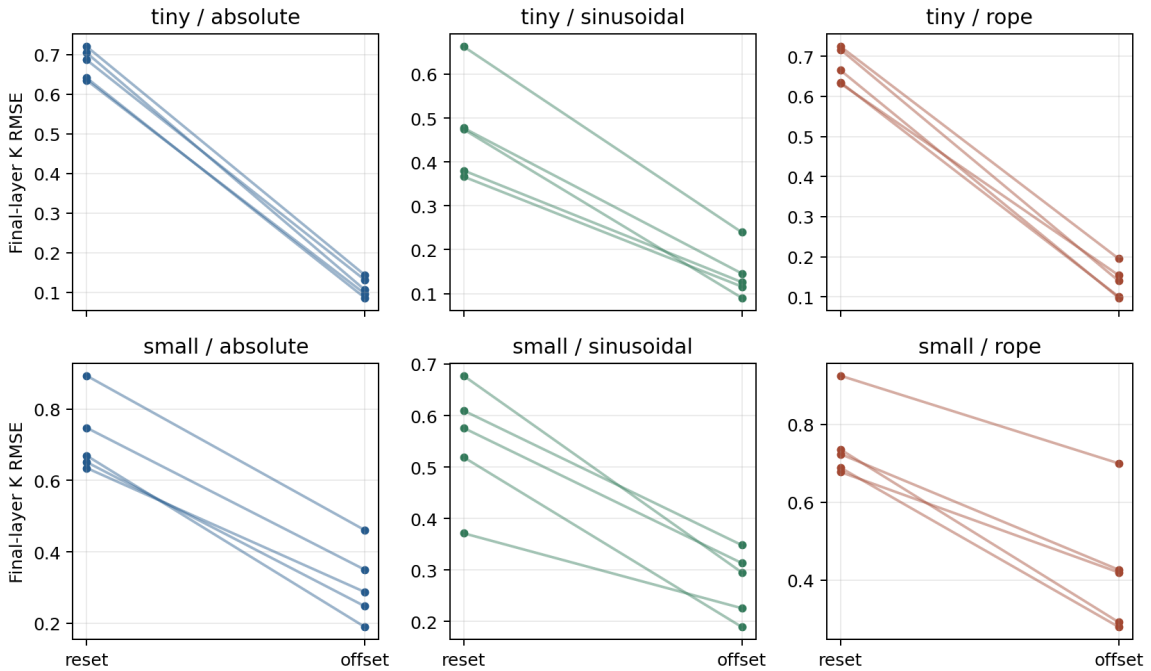


Figure 6: Final-layer reset-to-offset changes for every seed. Direction agreement is complete across two capacities and three positional mechanisms; residual error reflects missing block context rather than coordinate reset.

7.2.2 Continuous implicit-head progression

The capacity head probe advances one logical coordinate system through rollover. Every model call contains at most 24 tokens, below $L_{\text{model_top}} = 32$, while source positions span $L_{\text{logical}} = 192$; native-limit violations are zero. Learned-absolute IDs remain within $L_{\text{position}} = 192$. Table 7 shows that representation repair does not imply uniform end-task behavior.

Table 7: Continuous-head answer loss. Cells are mean / median over four examples and five seeds.

Tier	Position	Reset routed	Offset routed	Offset + overlap	Constructed oracle
Tiny	Absolute	2.3837/2.0986	.7304/.4736	.6474/.2631	.1384/.1136
Tiny	Sinusoidal	2.9691/2.5416	.9388/.0206	.8021/.0207	.0157/.0159
Tiny	RoPE	2.5910/1.5186	1.5547/.9828	.8763/.1163	1.0124/.6812
Small	Absolute	3.8306/2.2442	.6681/.0587	.0030/.0027	.0022/.0020
Small	Sinusoidal	5.0421/6.1229	3.3860/4.0264	.9110/.4171	.0011/.0011
Small	RoPE	6.7887/9.2938	6.2249/8.5253	3.0396/1.7279	.0012/.0011

Offsets lower seed-mean loss in 5/5 tiny absolute, 4/5 tiny sinusoidal, 3/5 tiny RoPE, and all 15 small-model pairs. Overlap lowers 2/5, 3/5, and 5/5 tiny seeds and 4/5, 4/5, and 5/5 small seeds in absolute/sinusoidal/RoPE order. Evidence-chunk recall generally rises after offsets, but remains mechanism- and seed-dependent before the oracle. The constructed oracle sets recall to one and nearly eliminates all small-model loss. It is not universally superior: tiny-RoPE mean loss worsens from 0.8763 to 1.0124. A single selected evidence chunk can omit useful surrounding context, so this failed preregistered expectation identifies composition as a separate factor rather than refuting the position contract. The large mean/median gaps in Table 7 also make clear why paired traces matter at five seeds.

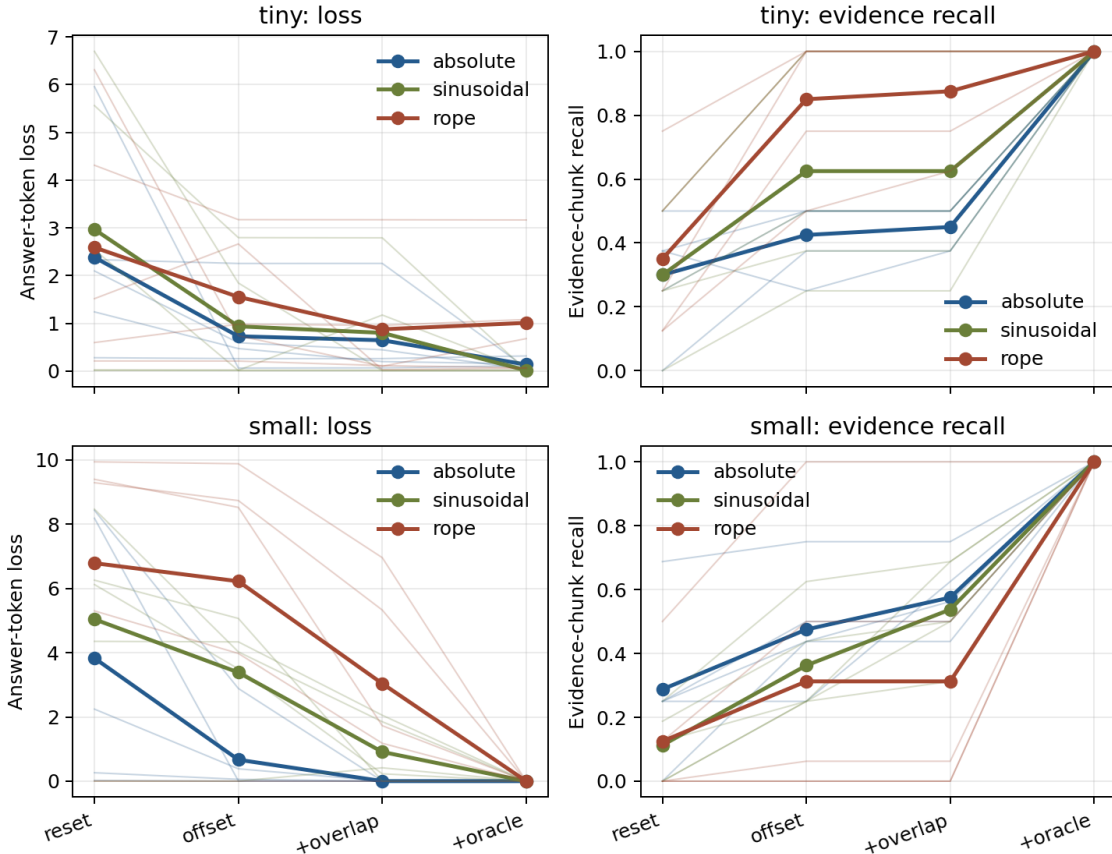


Figure 7: Continuous-head loss and evidence-chunk recall. Paired seed traces expose variation hidden by means; the separately recorded operation bound is independent of logical span.

7.2.3 Earlier distance-scaled implicit-head probe

All 1L–8L runs obey the 192-token native-operation limit; violations are zero. The largest call is 160 tokens at 1L and 192 thereafter. Table 8 gives mean and median over five seed means, each computed from four examples.

Table 8: Implicit-head answer loss. Each cell is mean / median over five seed means.

Tier	Position	1L	2L	4L	8L
Tiny	Absolute	.143/.104	.805/.920	1.338/1.382	1.206/1.403
Tiny	RoPE	1.043/.734	.543/.017	.830/.032	1.023/.565
Small	Absolute	.002/.002	1.520/.002	1.524/.004	1.618/.005
Small	RoPE	.001/.001	.062/.013	.129/.011	.282/.042

Small absolute means are dominated by one catastrophic seed; four of five seeds remain near zero. RoPE avoids that catastrophe but is lower in only two of five paired seeds at 2L–8L. Tiny results also change direction by seed and distance. This is reduced catastrophic variance in this sample, not a robust quality win. RoPE memory benefit over its own disabled condition is positive at 2L and 4L but negative at 8L: positional stability can coexist with unhelpful retrieval.

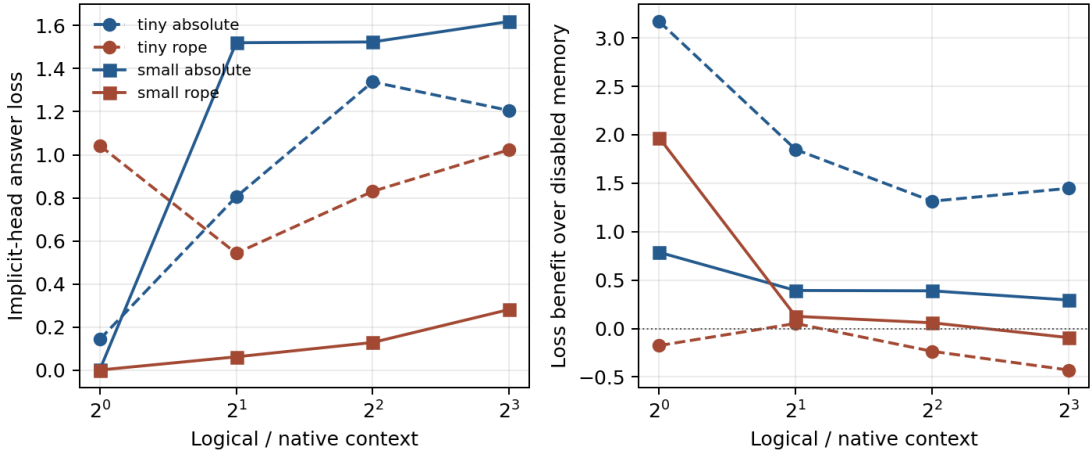


Figure 8: Bounded-head quality and benefit over disabled memory. Mean curves must be read with Table 8 because seed outliers are large.

7.3 Phase III—natural-language validation

At 8L, source-relative offsets reduce final-layer K RMSE in all 30 WikiText pairs, including all ten sinusoidal pairs. They do not produce a corresponding loss improvement (Table 9). Learned-absolute loss rises by 0.0200 in the tiny tier and 0.0292 in the small tier; sinusoidal loss rises by 0.0040 and 0.0707. RoPE loss is unchanged to the shown precision. Only 5/30 paired loss differences favor offsets, even though all 30 representation differences do. The preregistered expectation that offsets would often improve natural-text loss is therefore not supported at 8L.

Overlap offers a more specific result. It lowers final-layer K error further in every tier–mechanism condition and lowers loss in 29/30 paired runs relative to offset-only encoding. At 8L it processes

1.9375 times the unique source tokens. The loss reductions are modest, from 0.0007 to 0.0173 in the tiny tier and 0.0102 to 0.0148 in the small tier. Coordinate repair and contextualization repair are measurable, but neither should be equated with a large next-token quality change in these short-trained models.

Table 9: WikiText-2 at 8L. Deltas are intervention minus baseline, so negative is better. Each row contains five paired seeds.

Tier	Position	Δ final-K RMSE	Δ offset loss	Seeds better	Δ overlap loss	Seeds better
Tiny	Learned absolute	-.4281	+.0200	0/5	-.0173	5/5
Tiny	Sinusoidal	-.1897	+.0040	1/5	-.0007	4/5
Tiny	RoPE	-.6292	-.0000	2/5	-.0048	5/5
Small	Learned absolute	-.4044	+.0292	0/5	-.0140	5/5
Small	Sinusoidal	-.1810	+.0707	0/5	-.0102	5/5
Small	RoPE	-.6314	-.0000	2/5	-.0148	5/5

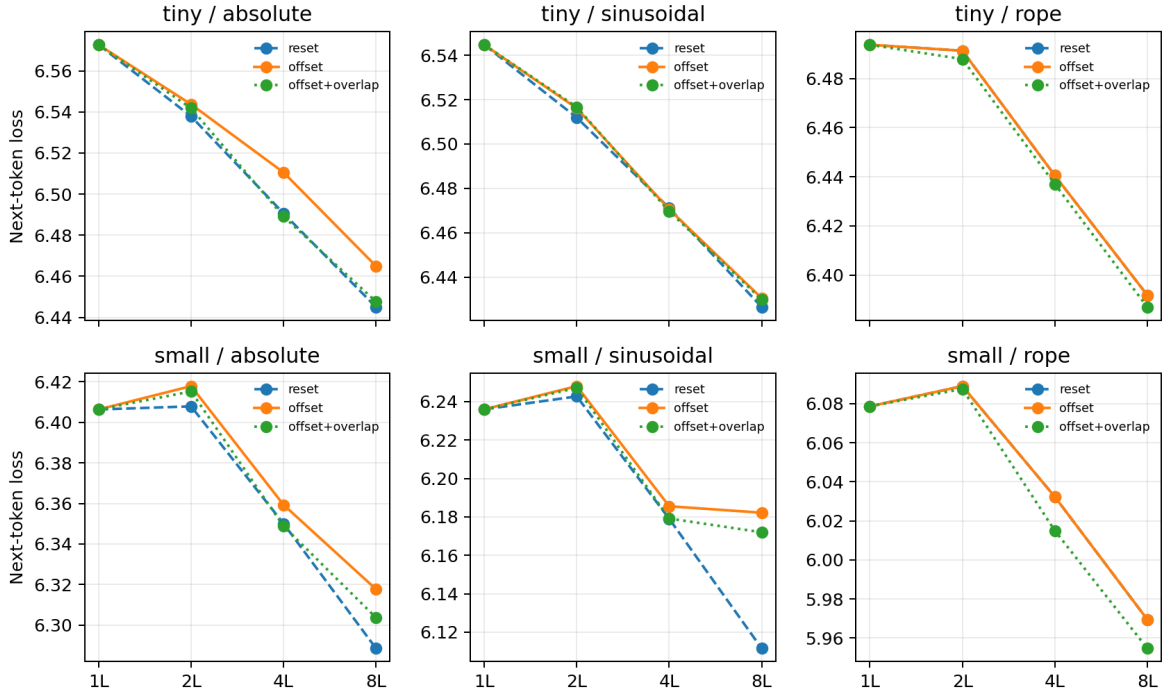


Figure 9: Bounded WikiText-2 next-token loss from 1L to 8L. Source offsets sharply improve representation fidelity but are neutral or adverse for loss; overlap gives small, more consistent improvements at the cost of duplicated encoding.

7.4 Phase IV—retrieved-memory validation

The controlled HotpotQA- and QASPER-derived probes produce a mechanism-dependent result (Table 10). Learned-absolute and sinusoidal routed loss improve on average in both tiers and datasets. RoPE routed loss instead worsens in all four dataset-tier cells. Across the full matrix, 35/60 paired seed effects favor offsets. This does not contradict the mechanistic K result: routing scores, relative source-query distances, distractor composition, and short-context training can change task utility even when the stored key has a more faithful source coordinate.

Table 10: Split-5 routed answer-token loss. “Better” counts paired seeds for offset versus reset; lower loss is better.

Dataset	Tier	Position	Reset	Offset	Offset + overlap	Better
HotpotQA	Tiny	Learned absolute	.559	.482	.482	4/5
HotpotQA	Tiny	Sinusoidal	.359	.304	.358	3/5
HotpotQA	Tiny	RoPE	.948	1.056	1.080	2/5
HotpotQA	Small	Learned absolute	.245	.043	.068	3/5
HotpotQA	Small	Sinusoidal	.061	.017	.012	4/5
HotpotQA	Small	RoPE	.129	.170	.196	1/5
QASPER	Tiny	Learned absolute	.629	.309	.301	4/5
QASPER	Tiny	Sinusoidal	.375	.244	.244	4/5
QASPER	Tiny	RoPE	.683	.869	.871	0/5
QASPER	Small	Learned absolute	.460	.299	.298	4/5
QASPER	Small	Sinusoidal	.403	.003	.007	5/5
QASPER	Small	RoPE	.169	.324	.184	1/5

Under offset plus overlap, answer-code accuracy ranges from 0.517 to 1.000 and target-chunk recall from 0.700 to 1.000 (Table 11). Median RCB is used because the mean is unstable when tail and full losses nearly coincide. MRR remains below recall, as expected when the correct chunk is selected but not ranked first. The physical materialization is about 50–54 token K/V states despite selecting roughly 2.8–3.6 routing chunks. The largest native operation in any HotpotQA/QASPER row is 53/52 tokens, respectively, below the configured 256-token limit; violations are zero.

Table 11: Split-5 routed metrics after offset plus overlap. RCB is the median; other values are means over examples and five seeds. KV is the number of materialized physical token states.

Dataset	Tier	Position	Loss	Accuracy	RCB	Recall	MRR	KV
HotpotQA	Tiny	Abs.	.482	.917	.975	.900	.696	51.6
HotpotQA	Tiny	Sin.	.358	.850	.924	.850	.588	51.5
HotpotQA	Tiny	RoPE	1.080	.517	.350	.700	.494	51.7
HotpotQA	Small	Abs.	.068	.967	1.000	1.000	.752	51.0
HotpotQA	Small	Sin.	.012	1.000	.998	1.000	.850	50.1
HotpotQA	Small	RoPE	.196	.917	.996	.950	.642	51.4
QASPER	Tiny	Abs.	.301	.900	.993	.983	.693	53.3
QASPER	Tiny	Sin.	.244	.933	.974	1.000	.776	53.1
QASPER	Tiny	RoPE	.871	.617	.295	.750	.501	54.4
QASPER	Small	Abs.	.298	.917	.999	1.000	.722	53.1
QASPER	Small	Sin.	.007	1.000	.999	1.000	.870	52.9
QASPER	Small	RoPE	.184	.933	.978	.983	.609	53.5

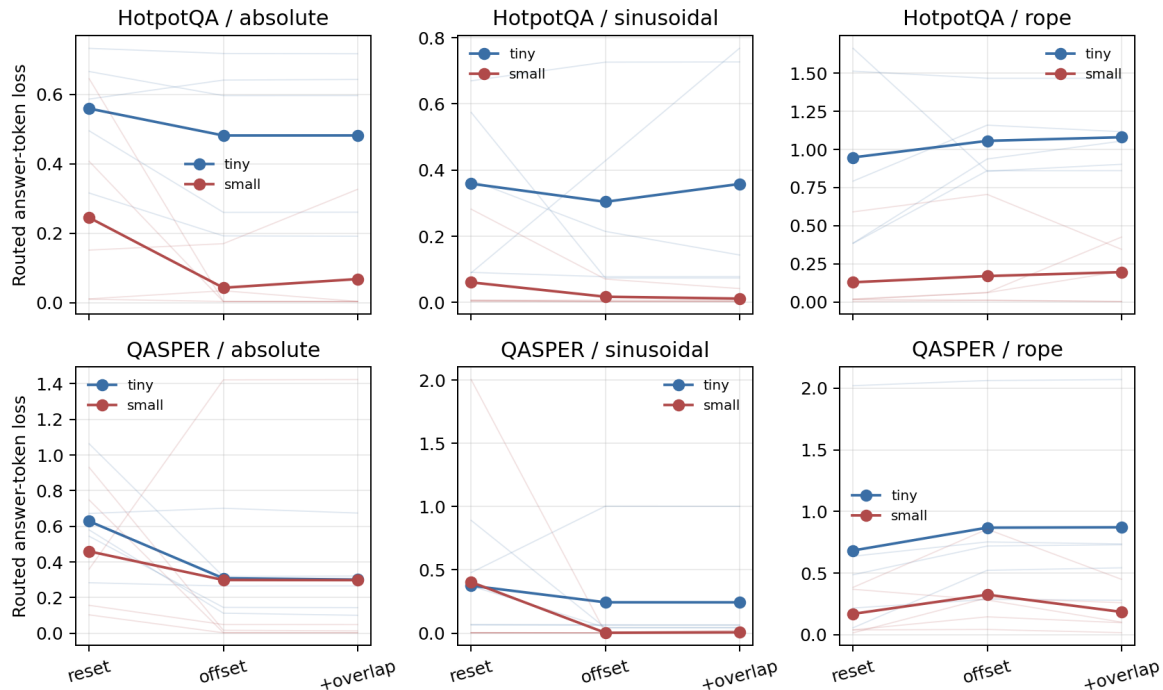


Figure 10: Routed loss under reset, source-relative offsets, and offsets plus overlap. Faint lines are individual seed means. Absolute and sinusoidal offsets improve on average; the RoPE direction reverses, and overlap is not monotonic.

7.5 Routing Recall Versus Realized Memory Fraction

The preceding fixed-selection recall reports whether the union executed by the multilayer router contains the programmatic target. The stored per-layer rankings permit a complementary retrospective: for each example and layer, select $k_i(f) = \lceil fN_i \rceil$ references and measure target-reference recall before cross-layer union. This is a routing experiment with labels; the evidence-only, all-memory, constructed oracle, and fixed-selection positional controls are deliberately excluded.

These probes contain only four candidate references. Consequently, requested fractions from 1% through 25% all realize one selected reference (25%), while 30% realizes two (50%). Table 12 retains the requested-fraction columns for consistency, but Figure 11 uses the honest realized fraction. f_{80} and f_{90} likewise report realized selection after ceiling. AUC_{0-30} integrates the predeclared requested-fraction grid and is included only as a compact within-protocol summary; it should not be mistaken for evidence below the 25% physical operating point.

Table 12: Small-tier offset-plus-overlap per-layer target-reference recall. R@5/10/20 all map to one of four references; R@30 maps to two. Inverse fractions are realized, not requested. Each row averages 240 layer rankings from five seeds.

Dataset	Position	R@5%	R@10%	R@20%	R@30%	f_{80}	f_{90}	AUC_{0-30}
HotpotQA	Learned absolute	.629	.629	.629	.721	1.00	1.00	.626
HotpotQA	Sinusoidal	.779	.779	.779	.829	.50	1.00	.770
HotpotQA	RoPE	.454	.454	.454	.588	1.00	1.00	.458
QASPER	Learned absolute	.571	.571	.571	.700	1.00	1.00	.572
QASPER	Sinusoidal	.800	.800	.800	.875	.25	1.00	.793
QASPER	RoPE	.408	.408	.408	.558	1.00	1.00	.414

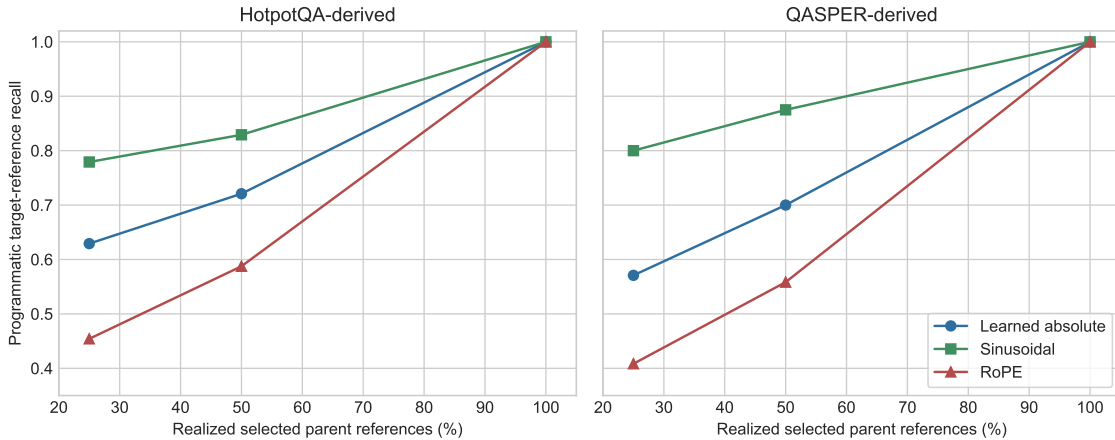


Figure 11: Per-layer target-reference recall at the three realizable selected-reference fractions. Sinusoidal routing dominates learned absolute and RoPE in this controlled probe. The four-candidate design cannot resolve behavior below 25%.

The ranking result is mechanism-specific rather than a generic benefit from coordinate repair. At the 25% operating point, sinusoidal recall is 0.779 on HotpotQA-derived and 0.800 on QASPER-derived text. Learned absolute reaches 0.629/0.571, while RoPE reaches 0.454/0.408. Moving to 50% raises all six conditions but leaves the ordering unchanged. The result agrees with the loss direction in Table 10: exact RoPE coordinate fidelity does not make the mean-gist router semantically stronger.

Exact candidate spans also permit a native-K/V axis. One selected reference materializes about 25% of indexed source K/V on average, two materialize about 50%, and all four give the complete endpoint. Figure 12 uses those physical token counts rather than treating references as equal length. The close agreement between the two x-axes is a property of this four-part generator, not a general PRA assumption.

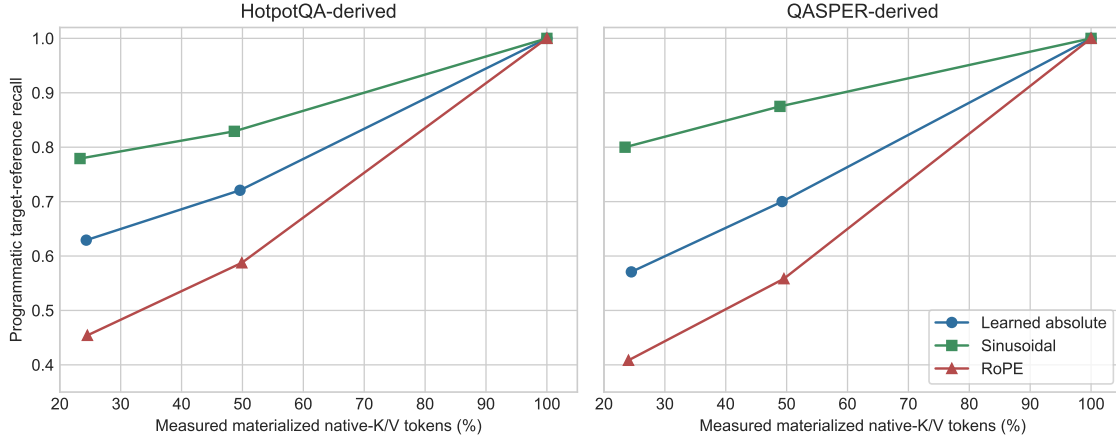


Figure 12: Per-layer target-reference recall against exact materialized native-K/V token fraction. Candidate lengths come from stored chunk spans; no token fraction is approximated.

Selection remains visible after coordinate repair, and the oracle anomaly suggests composition as a fourth candidate factor. The evidence-only condition is position-insensitive here because the explicit answer-code anchor occupies the first source chunk. All-memory loss improves under offsets for learned absolute and sinusoidal models but worsens for RoPE on average. At the example level, the nominal evidence oracle is worse than the router-selected set in 3.3%–78.3% of cells, depending on dataset, tier, and mechanism; conversely, mean routed loss usually exceeds evidence-only loss. Evidence identity therefore does not define a universally optimal memory composition. This is suggestive rather than a universal result. The inexpensive composition probe compares evidence-only, all-memory, routed, and shuffled sets; evidence plus exactly one irrelevant chunk and exhaustive best-subset search remain untested.

8 Discussion

What the position contract solves. Source-relative offsets, not RoPE alone, restore first-layer coordinate identity. Learned absolute positions do so within their table capacity; sinusoidal positions show that formula-defined absolute coordinates provide the same mechanical repair; RoPE additionally provides common-translation and explicit Q/K rebinding semantics. The paired direction survives the small tier. The continuous-head experiment confirms that a logical span can advance while every native operation remains bounded.

Representation fidelity versus utility. Correct coordinates recover positional fidelity but do not reconstruct missing causal context. Logical offsets attack positional fragmentation; overlap or historical encoding attacks the separate contextualization fragmentation caused by independently bounded blocks. The contract also does not choose the relevant chunk, prevent harmful attention to irrelevant K/V, or guarantee extrapolation from short training contexts.

Faithful K/V representation does not guarantee useful prediction: routing, learned distance distributions, and memory composition remain separate candidate determinants of task performance. At 8L, offsets improve WikiText final-layer K error in 30/30 pairs but improve loss in only 5/30. Retrieved-memory routed-loss effects are mixed across 60 pairs and reverse for RoPE means in both datasets and tiers. Routed-oracle gaps establish selection as an independent limitation; oracle anomalies suggest, but do not universally establish, memory composition as a fourth factor.

What is general and what is RoPE-specific. The evidence supports a narrower thesis than an across-domain quality claim. Source-relative positional continuity is a representation-fidelity contract for chunked native-KV transport across learned-absolute, sinusoidal, and RoPE models. RoPE’s distinctive contribution is not the offset itself; it is an explicit relative Q/K algebra that permits retrieval-time rebinding when raw K is retained. Once reset error is removed, residual task behavior can be attributed more cleanly—though not uniquely—to missing causal context, learned distance distribution, selection, and the candidate composition factor.

Post- versus pre-position K. Post-position K is immediately compatible with attention but binds the cache to source positions. Pre-position K is portable and compact offset metadata is small, but deferred rotation adds materialization work and requires a distance policy. Here it is 3.62 times the 160-token combined warm microbenchmark and does not reduce cache bytes. Exact logical assignment is faithful; compressed, bucketed, and order-preserving relocation change attention enough that their task quality remains an empirical question.

Engineering implication. Source identity, logical range, position state, and positional mechanism belong in cache provenance even when K is post-position. Without them, a consumer cannot audit local versus logical coordinates, reproduce a score, or safely introduce deferred rotation. For contiguous chunks, one logical start is enough for execution; full IDs can be reconstructed for traces.

9 Limitations and Threats to Validity

These are small from-scratch controlled models, not pretrained LLMs. The synthetic character tokenizer, binary answer-code task, 12 reference examples, four head examples, limited WikiText training, and GTX 950M limit external validity. Five seeds permit paired direction checks but

weak exact significance; an all-same-direction effect still has two-sided $p = 0.0625$. Mean loss is sensitive to rare catastrophes. The older split benchmark uses independent references with effective local positions and should not be read as the continuous-head source-offset result. The constructed oracle assumes the first evidence chunk and can omit useful context; it is a diagnostic intervention, not known optimal selection. Pre-position storage is not the canonical runtime path. We do not report production throughput, large-model memory savings, unrestricted natural-language QA, or streaming quality. The HotpotQA/QASPER sources are natural, but explicit answer-code supervision simplifies the target and changes the original task. The compact composition probe does not test every small subset. Streaming tests establish bounded rollover mechanics for all three modes, not useful generation beyond the native horizon. Formula-defined positions beyond the training range are mechanically valid but their quality is empirical. The distance-policy probe uses random content and must not be read as a language-model accuracy comparison.

10 Reproducibility

The pre-change baseline is tagged `paper1-pre-rope-baseline`; development occurs on `research/paper1-5-rope`. Earlier model training and split-quality measurements use `2b50f34cbf9f`; the original decomposition and storage matrix use `d36f67752701`. The final validation evaluators use `64f7cdf89b55`, and the generated cross-domain summaries use `052ae05`. Each JSON/CSV records its full source SHA, CUDA/PyTorch environment, seeds, and protocol metadata. Raw JSON/CSV, plots, expectations, training histories, and the validation manifest live under `docs/papers/shared/results/paper1_5_rope/`; resumable checkpoints are under `out/paper1_5_rope/`. Tests cover translation, norm preservation, large logical offsets through overlap and splitting, independent absolute operation/position bounds, cache metadata, exact no-memory SA-to-PRA conversion, policy order, causality, deferred rotation, streaming rollover, and CPU/GPU parity. Existing Paper 1 tests remain the absolute-mode guardrail.

11 Conclusion

Source-relative positional continuity is required to faithfully preserve the intended positional identity of chunked native-KV memory, but faithful transport alone does not guarantee improved prediction. A compact source offset removes first-layer reset error for learned absolute, sinusoidal, and RoPE models, and the final-layer direction survives a small-model capacity increase. RoPE additionally supports deferred binding and common translation, but the unfused raw-K path is slower and K-only displacement deliberately changes attention.

Correct coordinates do not reconstruct missing causal context, and faithful K/V does not ensure task utility. The 30/30 representation improvements beside only 5/30 WikiText loss improvements and mechanism-dependent HotpotQA/QASPER effects locate the remaining problem in contextualization, learned distance distributions, routing, and a suggested memory-composition factor. This completes the paper’s planned controlled scope. Pretrained RoPE/GQA evaluation is the smallest next external-validity test, not another expansion of the present benchmark matrix. The routing retrospective also shows why that next study needs substantially more than four candidates: the current data has only 25%, 50%, and 100% realized operating points and cannot support a claim about highly sparse retrieval.

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